LEVELHULL · TRACEABILITY MATRIX

LevelHull traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/levelhull		
Source	docs/03b-traceability.md @ e0fd582		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.2	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.1	2026-10-25	Amish Chadha	Generated by .kit/trace.py

LevelHull traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 16 user requirements, 17 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Afloat when swamped	LVH-CAL-001, LVH-BLD-001	Analysis (LVH-CAL-001), TRL 3; swamp test based on the USCG level flotation method, TRL 4	Met on paper: carries 2.45 times the design load at 50 mm freeboard
UR-002	PR-002	Freeboard when swamped	LVH-CAL-001, LVH-BLD-001, LVH-STM-001	Analysis (LVH-CAL-001), TRL 3; measured in the swamp test, TRL 4	Met on paper in the free-flooding condition: 92 mm lowest, 113 mm amidships; 75 mm with timber at 1,100 kg/m3; brim-full straight after swamping a sealed hull floats about 24 mm level and about 3 mm at the lowest point until the crew bail (LVH-STM-001 S2, S32)
UR-003	PR-003	Floats level	LVH-CAL-001, LVH-BLD-001, LVH-STM-001	Analysis (LVH-CAL-001), TRL 3; inclinometer in the swamp test, TRL 4	Met on paper with the water inside flowing freely: 0.0 deg heel, 0.39 deg trim; brim-full a sealed hull is near neutral in roll and lolls a degree or two until it spills (LVH-STM-001 S33, to confirm by test)
UR-004	PR-004	Bail-out	LVH-CAL-001, LVH-BLD-001, LVH-STM-001	Analysis (LVH-CAL-001, estimate), TRL 3; timed trial, TRL 4	Met on paper (estimate): about 11 min from brim-full at a sustained 50 L/min each (LVH-STM-001 S34); about 3 min from the lake-level condition at 100 L/min each; rate to measure in the trial (S17)
UR-005	PR-005	Working space	LVH-CAL-001, LVH-BLD-001	Analysis (LVH-CAL-001), TRL 3; measured on the fitted canoe, TRL 4	Met: 8.4 %
UR-006	PR-006	Water uptake	LVH-CAL-001, LVH-STM-001	Data sheet (LVH-STM-001 S8), TRL 3; immersion and weighing of samples, TRL 4	Met on supplier data (under 2 % by volume, about 2 % of lift); to confirm by test
UR-007	PR-007	Fixing strength	LVH-CAL-001, LVH-BLD-001, LVH-STM-001	Analysis (LVH-CAL-001, LVH-STM-001 S11 to S13), TRL 3; CalRig proof load of a batten, screws and strap, TRL 4	Met on paper: lowest factor 5.0 (coach screw); chock screws now 31 mm into the batten
UR-008	PR-008	Fitting time		Analysis (LVH-CAL-001, estimate), TRL 3; timed fitting	Met on paper (estimate): about 6.7 h

User requirement	Product requirement	Title	Evidence	Verification	Status
			LVH-CAL-001, LVH-BLD-001, LVH-STM-001	with the partner builder, TRL 4	
UR-009, UR-006	PR-009	Durability	LVH-CAL-001, LVH-STM-001	Field inspection of pilot canoes, TRL 5 and later	To verify at TRL 4: cannot be shown on paper (LVH-STM-001 S10, S23)
UR-010	PR-010	Cost	LVH-CAL-001	Costed bill of materials, TRL 3	Over target: USD 1,384, over the value-engineering target by USD 384 (four bailers per canoe since D-O3)
UR-011, UR-001	PR-011	Reserve lift	LVH-CAL-001, LVH-BLD-001	Analysis (LVH-CAL-001), TRL 3; swamp test with added weights, TRL 4	Met on paper: 2.45
UR-012	PR-012	Damage tolerance	LVH-CAL-001, LVH-BLD-001, LVH-STM-001, LVH-OPT-001	Analysis (LVH-CAL-001), TRL 3; swamp test with one module removed, TRL 4	Met on paper: 25 mm, 1.5 deg heel
UR-013	PR-013	Crew on one side	LVH-CAL-001, LVH-BLD-001	Analysis (LVH-CAL-001), TRL 3; swamp test, TRL 4	Met on paper: 3.9 deg (low-side freeboard 44 mm)
UR-014, UR-005	PR-014	Inside the hull lines	LVH-CAL-001	Inspection of the model and drawing (LVH-DWG-001), TRL 3; inspection of the fitted canoe, TRL 4	Met by design: modules, battens and straps inside the hull; grab lines and eyes stand about 45 mm outside the strake
UR-015	PR-015	Life jacket message	LVH-CAL-001	Inspection, TRL 3 (documents) and TRL 4 (kit marking)	Met by design for the documents; kit marking to add with the freeboard marks at TRL 4
UR-016	PR-016	Sizing calculator	LVH-CAL-001	Inspection of `docs/04-calcs/sizing.py` (LVH-CAL-001 section 8), TRL 3; check against measured hulls, TRL 4	Met by design: sizing sheet for 6.5, 8.0 and 10 m canoes
UR-008	PR-017	No holes below the waterline	LVH-CAL-001	Analysis on the model (LVH-STM-001 S19), TRL 3; inspection, TRL 4	Met on paper: only the eight eye bolt holes, 450 mm above the 175 mm loaded waterline; coach screws stop 12 mm short of the planks

Gaps

None.